Abhay Gupta

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Portfolio: abgup.com & notes.abgup.com
Current: Seattle, WA
U.S. Citizen

*Resume coded in **LaTeX***

PUBLICATIONS, CONFERENCES, AND PATENTS

Automobile Damage Detection Using Thermal Conductivity, Filed: Dec 2018

- Inventors: J. Schow, and **A. Gupta**
- ***U.S. Patent*** #: 11,580,791 Feb 2023

A Cellular Automaton for Modeling Non-Trivial Biomembrane Ruptures, Submitted: Jul 2018

- **A. Gupta**, G. Reint, I. Gozen, and M. Taylor
- Presented at ***The 13th World Congress in Computational Mechanics (WCCM)*** Sep 2018
- Session: *Novel Mathematical Models and Computational Methods*, New York, NY
- Published in ***Soft Matter*** Apr 2019

SKILLS

Languages
- Python & Lua
- C# & Java
- Javascript & VBA
- HTML/CSS
- Bash & Powershell
- Vimscript
- C/C++
- LabView

Frameworks
- Django
- WinForms/WPF
- Flask
- Jekyll
- Cordova
- JQuery
- Next.js
- LaTeX & Maple
- Matlab/Simulink

Software Tools
- Git
- Vim/Neovim
- Cmake
- Jira/Azure
- Docker
- Figma
- Microsoft Office
- Linux
- Android/iOS

Modelling/Analysis
- SolidWorks
- Siemens NX
- OnShape/Auto-CAD
- Ansys CFX
- Star CCM+
- Abaqus
- FreeCAD
- Fusion 360

Hardware Tools
- 3D Printers
- Lathe/Mill
- Drill Press
- Bandsaws
- Sanders/Grinders
- Oscilloscopes
- Soldering
- Function Generators

EDUCATION

University of Washington, M.S. Robotics & Data Science 2018-2019

- Thesis: 3D Graphics & Regression Models || 3.6
- Advisors: Steve Brunton & Ashis Banerjee
- Taught graduate mathematics courses for engineering
- Researched & developed a custom physics simulation of Optical Tweezers and digital twin models for compliant motors via machine learning regression methods

  Python || MATLAB || C++ OpenGL || ROS

Santa Clara University, M.S. Computer Engineering 2017-2018

- Transferred to University of Washington || 3.8
- Simulated biological membrane fracture through AI methods (see paper in publications)

  C || C++

Santa Clara University, B.S. Mechanical Engineering 2014-2017

- Entrepreneurship minor || Graduated in 3 years with honors || 3.6
- Assisted teaching course in Numerical Analysis to undergraduate engineering students

  MATLAB || LaTeX || Maple Simulink || LabVIEW SolidWorks || Abaqus || Star-CCM+ Oscilloscopes || Lathe || Mill

INDUSTRY EXPERIENCE

Pebble Products LLC, Founder Aug 2022 - Present

- Develop/prototytpe new hardware/software products
- Contract engineering services to local companies/individuals

  notes.abgup.com pebbleproducts.com

Kawaskai Robotics, Software Engineer, Robotics 2022

- Developed software to move wafer handling (scara) robots
- Tested physical robotics arms on local and vendor facilities to ensure functionality
- Optimized for throughput and reach requirements to maximize computer chip production capability

  C language AS (DSL)

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SummerBio, **Software Engineer, Robotics** 2021 - 2022

- Developed software drivers (6 DOF arms, benchtop systems) for VWorks
- Built communication platforms through both Ethernet and serial port protocols
- Built hardware testing rigs to ensure new hardware/drivers work with automation line
- Unit tested drivers through manual isolation testing and automated via C# test framework, xUnit

C# || Python || VBA   Figma

Lam Research, **Software & Mechanical Engineer** 2020 - 2021

- Led and assisted Android app development via Cordova (JS) framework
- Developed/Maintained software to optimize a 6 DOF robotic arm via C# WinForms
- Built communication protocols for hardware components via Modbus, Ethernet, and Bluetooth protocols
- Physically tested software and hardware in clean room environment via systematic approach
- Analyzed test results via data science techniques to ensure repeatability and reliability metrics
- Led development of and maintained a fishbone diagram issue diagnosis application with offshore developers via C# Xiamarin Framework on Windows and Android
- Automated Android content upload and verification via VBA excel automation tools
- Worked alongside machinists to develop custom test rigs and hardware components (sheet metal, aluminum/steel parts, injection molded designs, etc) for robotics arm applications
- Designed and printed custom 3D components on a MakerBot and local venders
- Designed a custom electronics cart via Siemens NX fitting custom components and facility constraints

C# || JavaScript || Python   NX || Teamcenter   Figma   VBA

CSAA Insurance, **Physics Consultant** 2018 - 2020

- Evaluated novel engineering and physics aspects for patent applications
- Researched alternative approaches for products and methods

INTERNSHIPS

Microvision, **SDE Intern** Summer 2019

- Modeled the response of Lidar activated SiPM (Silicon photomultipliers)

Simulink || MATLAB   LTSpice

TheraNova, **SDE Intern** Summer 2018

- Developed a python software analysis system to understand gait measurements
- Ensured the software analysis is accurate for 80+ patients

Python || MATLAB

Valeo, **Systems Engr Intern** Spring 2018

- Produced hardware and software demos for automotive OEMs
- Collaborated with start-ups and OEMs to develop new cabin safety features

VBA

Pentair, **ME/EE Intern** Fall/Winter 2017

- Optimized performance of steady state and transient phases of circuit breakers
- Laboratory tested and simulated rail heating to melt snow

Ansys CFX || SolidWorks   Oscilloscopes, Function Generators   DC/AC Power Supplies   Soldering

Accel Biotech, **ME Intern** Summer 2016

- Prototyped medical device components and test assemblies
- Supported mechanical, electrical, and software design of a blood diagnostic device

SolidWorks, 3D Printing   Oscilloscopes

Caltrans, **ME Intern** Summer 2015

- Reviewed and advised on structural testing for next generation locomotives
- Designed a floor plan using Microsoft Visio & participated in vendor meetings

Microsoft Visio

CERTIFICATIONS

Engineering-In-Training, **State of CA** May 2018

Solidworks CSWA, **Dassault Systèmes** Mar 2015

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VOLUNTEERING

Auto Angels, Bellevue, **Car Mechanic** Jun 2024 - Present

Rainier Scholars, Seattle, **Computer Science Teacher** Jun 2024 - Present

The Bikery, Seattle, **Bicycle Board Member & Technician** Nov 2023 - Present

Capitol Hill Tool Library, Seattle, **Volunteer** May 2024 - May 2025

Santa Clara University, Alumni, **Engineering Mentor** Oct 2023 - Jun 2025

FIRST Robotics Competition (FRC) 254, **Engineering Mentor** Aug 2021 - Present

Reddit Group: r/ControlTheory, **Moderator** Dec 2018 - Present

San Jose Bicycle Clinic, **Bicycle Technician** Jul 2020 - Jan 2022

SOFTWARE PROJECTS

206bikepolo.com, **Web Developer**

- Update and maintain website using Javascript framework (Next.js)

  HTML || CSS || Javascript   Next.js || React   206bikepolo.com

SheldonBrown.com, **Web Developer**

- Build and maintain website to share bicycle repair information
- The most popular bike repair website worldwide

  HTML || CSS || Javascript   Wordpress

Find a Paint, **Web Developer**

- Developed website to compare paint brands to help high school mom find corresponding paint for work

  Python || HTML/CSS || JS   Flask   findapaint.com

Masala Blend, **Web Developer**

- Help stepmother build webpage to sell homemade spices locally

  HTML || CSS || Javascript   Shopify

Personal Website & Resume, **Web Developer**

- Build personal website via github pages & programmatically render resume via Latex

  HTML || CSS || Javascript   Jekyll   LaTex || LualaTex   abgup.com

Peer Portfolio Consulting, **Web Developer**

- Assist undergraduates build their resume & web porfolio to gain industry experience

  HTML || CSS || Javascript   Python || Latex   Django   github.com/carly85/CV

HARDWARE PROJECTS

Cap Hill Tool Library, **3D Printing Assistant** 2022-Present

- Assist community members and create models/prints for in-house repairs and personal projects

  Onshape || BambuLabs   Ender 3

Bike Polo, **Mallet Design/Fabrication**

- Model, 3D print, and fabricate custom bike polo mallet
- Iterate through design process, optimizing for weight and reduction of stress concentration/shattering

  Onshape   Drill Press || Bandsaw

LED Hula Hoop, **Design/Fabrication**

- Build custom LED timed lighting to match the circadium rhythm of the sun
- Change lighting within building from blue light to gradually hit red light throughout day

  Arduino || C/C++   Soldering

HOBBIES

Bicycling || Handstands || Basketball || Bike Polo || Unicycling || Tennis
Running || Skateboarding || Rollerblading || Dancing
Weight Lifting || Yoga
Board Games || Drawing || Reading || Cooking
Coding || DIY Projects || Electronics || 3D Printing/CAD || Fabrication || Robotics

For more: abgup.com